Dear Quillstone Payments AI Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. At Thistlebox, I build and evaluate machine-learning workflows that support operational and product decisions, including forecasting, churn modelling and experimentation. My weekly recipe-demand forecast uses LightGBM and engineered calendar, price and menu-position features; scheduling it through Vertex AI Workbench to write results to BigQuery helped improve WAPE from 23% to 14%, with an estimated 9% reduction in food waste.

I have also developed a weekly XGBoost churn-scoring workflow in BigQuery using dbt, and built a Streamlit calculator for planning and analysing sequential A/B tests, now used by around 25 colleagues in product and growth. I wrote the calculator's Dockerfile, and I began adding pytest unit tests to forecasting feature code in 2025, reaching approximately 60% coverage. My experience so far has focused on scheduled and batch workflows rather than serving models through an API; I am now looking to grow toward machine-learning engineering and deepen my experience building production systems.

Quillstone's focus on helping merchants and its close collaboration across AI, engineering, product and operations make this role particularly appealing. I would bring a strong foundation in Python, feature engineering, statistics and applied machine learning, along with experience translating analytical work into tools and workflows that colleagues can use. I hold an MSc in Statistics, with a dissertation on Bayesian hierarchical models, and have also completed a personal image-classification course project using PyTorch.

Thank you for considering my application. I would welcome the opportunity to discuss how my experience could contribute to Quillstone's Customer Operations AI team.

Kind regards,
Tobias Mwangi-Hartley